

ISHMAM HAQUE

iahaque2000@gmail.com | +1 3472834721 | <https://github.com/IshmamHaque1112> | [linkedin.com/in/ishmam-haque](https://www.linkedin.com/in/ishmam-haque)
EXPERIENCE

CUNY BUILDING PERFORMANCE LAB

Energy Codes Machine Reading Intern

July 2022 – August 2022

- Created computer vision algorithms with Python in Jupyter to capture and process blueprint images to text to analyze building energy efficiency, using Tesseract-OCR, sci-kit learn, pandas, imutils, and other libraries.
- Developed a Jupyter notebook that could process images of charts to an extent into data structures, then created six notebooks to improve issues with my original program such as spelling errors.

CODEPATH

Summer Technical Interview Teaching Assistant

June-August 2022 and 2023

- Facilitated sessions to educate students on prospective topics for technical interviews ranging from binary trees, to stacks and queues, with most of the content focused on Python.
- Managed several Zoom group pod sessions biweekly totaling around 20 students, utilizing my skillset to re-explain difficult concepts, offer coding recommendations, and conduct interviews.

NYC DEPARTMENT OF EDUCATION

Substitute Educational Paraprofessional

January 2022-present

- Supporting teachers and school staff through various activities such as by tutoring and advising students, ensuring discipline in class, and aiding special needs students.

PROJECTS

STORE SIMULATION - Queens College: Software Engineering

September–November 2021

- Captained a six-member team to develop a store simulation in Java in the agile approach, where one could log on and play the role of either a boss, an employee, or a customer.
- Leading a series of weekly coding sessions on Replit, delineated roles, designed and programmed over 30 classes, implemented design patterns for coding efficiency, and documented the project extensively.

MULTIPLAYER GAMING SERVER- Queens College: Data Communications

October-December 2022

- Teamed up with two colleagues to devise a multiplayer game server in Java, utilizing TCP threaded connections to play Crazy Eights, the dice game Farkle, or Trivia.
- Programmed Farkle by creating a playable demo and configuring the code to work with the client-server project code, and tested server connections to ensure functionality.

KNAPSACK PROBLEM SOLUTIONS- Queens College: Genetic Algorithms

February-May 2023

- Designed three Java programs solving the knapsack problem, where one must determine the best selection of items from a collection with maximum value under a weight limit.
- Created the first approach in brute force generating all possible solutions to find the best, second in random search generating 100 generations of 1000 random solutions for the final one.
- Developed the third approach in genetic algorithms; over each generation of 1000 solutions, top solutions breed to make the next generation, the final solution being the best in the 100th.

MUSHROOM POISONING PREDICTIONS- Queens College: Applied Data Science

December 2022

- Deployed Python, sci-kit learn and pandas to preprocess a 1988 mushroom dataset and train random forest, GNB matrix, and SVM models to find the best model and parameters to predict the most poisonous species.
- Delivered a presentation to explain how my best model (the random forest) could be utilized for daily research into the usability of mushroom species for practical purposes.

FICTIONAL COLLEGE DATABASE- Queens College: Database Systems

December 2021

- Participated in creating a fictional college SQL database by creating Departments and Instructors tables, defining type constraints and keys, and loading premade data with coded procedures.

PARALLEL PROCESSING ARRAY MINIMUMS, Queens College: Operating Systems

April 2021

- Determined the minimum of a 10,000-size array in C using parallel processing by partitioning into 1000-size arrays, sorting the local minimum of each array with p threads, then sorting the locals to find the global minimum.

CHAIN CODING, Queens College: Computer Vision

April 2023

- Designed a C++ program that accepts a text pseudo-image replacing bits for pixels, traces object boundaries in the image with chain coding, and outputs a text file only displaying boundaries.

SOCIAL MEDIA API IN SPRING, Revature

December 2024

- Developed a full stack social media blog API using Java, Json, and SQL allowing a user to do CRUD (Create, Read, Update, Delete) actions such as creating an account or message posting.
- Utilized the Spring framework in Java as a means of simplifying code by managing dependencies and configuring Json data connections with a web controller to handle CRUD operations.

EDUCATION

CUNY, Queens College—*B. S. in Comp Sci: Minor in Business & Liberal Arts*

Graduated Jun 2023

- **Cumulative GPA 3.52/4.0** | Honors in Cum Laude
- **Relevant Coursework:** Algorithmic Problem-Solving, Object-Oriented Programming, Theory of Computation, Software Engineering, Data Communications, Database Systems, Data Science & Applied Machine Learning

CUNY Tech Prep—*Data Science Fellow*

July 2021–Jun 2022

COOP Careers—*Data Analytics Apprentice (Cohort 340)*

Aug 2023–Dec 2023

Revature—*Entry Level Software Engineer (Unpaid Training, Assessment Passed)*

Oct 2024–Dec 2024

SKILLS

Technical: C/C++, Java, MIPS, Python 3, SQL, Swift, LISP, HTML, CSS, JavaScript, Eclipse, Ruby, PowerShell, XCode, Anaconda, Linux (SSH), SMARTYY, SSMS, Pandas/numpy, Jupyter, Scikit-learn, Tesseract-OCR, Windows, Tableau, Microsoft Office, Spring, Json

Certifications: Coursera Google Data Analytics, CUNY Tech Prep, CUNY TTP bootcamp, Codepath iOS Bootcamp